

Ritesh Tiwari

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ABOUT ME

Mathematics undergraduate at NIT Surat with research interests in real analysis, numerical computation, and mathematical visualization. Has a strong foundation in mathematical reasoning and programming, with a focus on applying theoretical concepts to computational and research-oriented projects.

EDUCATION

Integrated M.Sc. in Mathematics 2024–Present
Sardar Vallabhbhai National Institute of Technology Surat, Gujarat, India

- **Current CGPA:** 8.98
- **Related coursework:** Real Analysis, Calculus, Group Theory, Graph-Theory, Data Structures.

CLASS XII [Board: CBSE] 2023
Kendriya Vidyalaya, NERIST, Nirjuli, Arunachal Pradesh

- **Percentage:** 92.4%

CLASS X [Board: CBSE] 2021
Vivekananda Kendra Vidyalaya, P.T.C Banderdewa, Arunachal Pradesh

- **Percentage:** 94.8%

TECHNICAL SKILLS

- **Core Strengths:** Logical reasoning, mathematical thinking, analytical problem-solving
- **Programming languages:** C++, Python, MATLAB
- **Mathematical tools:** LaTeX(basic), Desmos
- **Python libraries:** Numpy, Matplotlib, Sympy (basic knowledge)
- **Other tools:** Jupyter Notebook, VS Code

COURSES & CERTIFICATIONS

- **Linear Algebra For Machine Learning [STTP]** – LDRP Institute of Technology & Research, GTU (Dec 2025)
- **Mastering LaTeX & Scientific Sketching** – IMPA & Infinity Research and Development Institute (Nov 2025)
- **Statistics & Data Analysis Workshop** – Sikkim University (Oct 2025)
- **SciComm 101 Workshop** – IISc Bangalore (Oct 2025)
- **Python for Linear Algebra and Numerical Methods** – IMPA & Infinity Research and Development Institute (Jul 2025)
- **MATLAB Course** – Internshala (May–Jul 2025)
- **Single Variable Calculus** – MIT OpenCourseWare (online)

PROJECTS

Project 1: Solving Ordinary Differential Equations Using Numerical Methods in MATLAB (May–July 2025)

- Implemented Euler's, Heun's, and Runge-Kutta methods for solving ODEs.
- Compared results with analytical solutions and MATLAB's built-in function `ode45`.
- Created graphical visualizations to compare the precision and convergence of each method.

INDEPENDENT STUDY

A Mathematical Study of Convergence of Selected Real Sequences

[Self-Initiated Project (2026)]

- Analytically studied convergence of selected real sequences using limit theorems and algebra of sequences.
- Verified convergence rigorously using the ε -N definition of limits.
- Performed numerical error analysis to compare rates of convergence.
- Created graphical visualizations to illustrate convergence behavior and monotonicity.
- Developed supporting theory using power series, Taylor expansions, and Ratio Test.

Tools Used : LaTeX, Python, Desmos

AWARDS

- Won **3rd prize** in the event **Maths Maze** held during **National Mathematics Week (16-23 dec, 2025)** at SVNIT, Surat.

INTERESTS

- Mathematical visualization
- Solving mathematical puzzles